

Federal University of Sergipe

Special Topics in SE and IS I

Prof. Glauco Carneiro · June 2026

MedEval — RAG-as-a-Judge Framework

Activity 3 | Group 5

Evaluation of Retrieval-Augmented Generation (RAG) and
RAG-as-a-Judge on Medical Language Models
using K-QA and USMLE Datasets

Team Member	Role
Moaath Almohammad Alshaikh	Full-stack dev., RAG architecture, pgvector, statistics
Tasneem Alshafer	SLM models, RAG re-inference pipeline, dataset management
Marcelo West	Database schema extension, backup
Clélio Xavier	Judge pipeline adaptation for RAG
Sérgio Santos	Human evaluation, before/after results analysis
Hernandison Bispo	Research insights, document curation

Repository

[https://github.com/moaathalshaikh/
MedEval-RAG-as-a-Judge-Framework-Atividade-3](https://github.com/moaathalshaikh/MedEval-RAG-as-a-Judge-Framework-Atividade-3)

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1 Overview

MedEval Activity 3 extends the LLM-as-a-Judge evaluation platform built in Activity 2 with a complete Retrieval-Augmented Generation (RAG) pipeline. The system ingests authoritative clinical guidelines published *after* the training cutoff of all evaluated candidate models, embeds them into a vector knowledge base, re-runs inference on the same medical QA datasets, and measures the factual impact of external knowledge through before/after Spearman rank-correlation analysis.

“A poorly selected document destroys the correlation coefficient of your model and will be penalized by your own judge agent.”

The central research question is: **does injection of recent, authoritative clinical knowledge improve factual correctness, reduce hallucinations, and increase alignment between LLM-judge scores and human evaluator scores?**

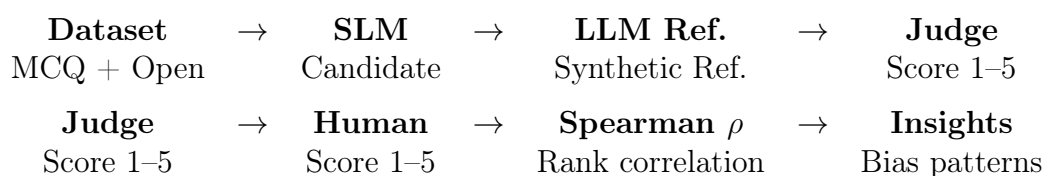
The full source code and replication package are publicly available at:

<https://github.com/moaathalshaikh/MedEval-RAG-as-a-Judge-Framework-Atividade-3>

2 System Pipeline

The evaluation pipeline transforms raw medical questions into actionable research insights through a multi-stage process combining automated inference with human-in-the-loop validation.

2.1 High-Level Flow



Note

The LLM-generated reference answer is a **Synthetic Reference** used to guide judge scoring. The true gold standard remains the original annotated answer from the dataset. Human-in-the-loop validation is the key differentiator from traditional automated metrics (BLEU, ROUGE).

2.2 RAG Architecture

The RAG component extends the Activity 2 pipeline with the following stages:

Step 1. Document Ingestion — Upload PDF/TXT → chunk (500 chars, 50-char overlap) → embed via `text-embedding-3-small` → store in pgvector (`rag_chunks`).

Step 2. RAG Re-Inference — For each question in K-QA / USMLE:

- a. Embed the question text.

- b. Cosine similarity search → top-5 relevant chunks.
- c. Inject chunks into the system prompt.
- d. Call the SLM → save response with `rag_enabled = true`.

Step 3. Judge Evaluation (RAG-aware) — Same judge model (DeepSeek-v4-flash), same 1–5 rubric. Additional Chain-of-Thought field: RAG HELPED / RAG ADDED NOISE / RAG NEUTRAL.

Step 4. Database Update — Populate `score_with_rag`, `rag_impact`, and `cot_rag` in `judge_evaluations`.

Step 5. Statistical Analysis — Compute Spearman ρ (judge vs. human) for baseline and RAG conditions; report $\Delta\rho$ per model and dataset; flag and export noise cases.

3 Evaluation Methodology

3.1 Judge Scoring Rubric (1–5)

Table 1: LLM-as-a-Judge scoring rubric for open-ended responses.

Score	Label	Criteria
5	Excellent	Fully correct, complete, and well-reasoned. Matches or exceeds the reference answer.
4	Good	Mostly correct with minor omissions. Clinically acceptable with small gaps.
3	Partial	Partially correct. Captures the main idea but misses key clinical details.
2	Weak	Mostly incorrect or incomplete. Shows limited understanding of the question.
1	Critical	Completely incorrect, irrelevant, or potentially harmful in a clinical context.

MCQ questions are graded deterministically: the model’s selected letter is compared against the gold answer.

3.2 Computing Human Average Scores

When multiple human reviewers evaluate the same response, the system computes the arithmetic mean of all submitted scores:

$$\overline{H} = \frac{s_1 + s_2 + \cdots + s_n}{n} \quad (1)$$

This average is then compared against the judge score to compute the disagreement delta (Δ) and to build the Spearman rank correlation.

3.3 Disagreement Detection Threshold

$$\text{Disagreement } |\bar{H} - J| \geq 2 \quad (2)$$

- **OVERRATING** ($J > \bar{H}$): The judge is more lenient than human reviewers. May indicate sensitivity to response style, verbosity, or lexical similarity with the reference.
- **UNDERRATING** ($J < \bar{H}$): The judge is stricter than human reviewers. May indicate failure in semantic understanding or paraphrase detection.

4 RAG Knowledge Base — Document Selection

Documents were selected according to three criteria:

1. **Temporal validity** — publication date *after* the training cutoff of all three evaluated candidate models (BioMistral-7B, Mistral-7B-Instruct-v0.1, Qwen1.5-1.8B-Chat; all with cutoff $\leq$ September 2023).
2. **Topical relevance** — direct alignment with K-QA and USMLE question distributions (infectious diseases, pharmacology, nephrology, cardiology, mental health).
3. **Clinical authority** — guidelines issued by internationally recognised bodies (CDC, IDSA, AHA/ACC, KDIGO, FDA, NIMH, USMLE.org).

Table 2: RAG knowledge base documents.

#	Document	Source	Year	Covers
1	STI Treatment Guidelines	CDC	2021	Chlamydia, Herpes, HPV, gonorrhea
2	Antimicrobial Stewardship	IDSA	2023	Antibiotic selection and stewardship
3	ACC/AHA Hypertension Guidelines	AHA Journals	2017/2023	Hypertension management
4	KDIGO AKI & AKD Guidelines	KDIGO	2023/2026	Acute Tubular Necrosis, AKI staging
5	FDA Drug Safety Communications	FDA	2024	Post-cutoff drug safety updates
6	USMLE Step 1 Content Outline	USMLE.org	2024	Biomedical sciences blueprint
7	Depression & Anxiety Clinical Info	NIMH	2024	SSRIs, buspirone, mental health Tx

5 Key Results

Table 3: Summary of experimental results (Activity 3).

Metric	Baseline (no RAG)	With RAG
Spearman ρ (Judge vs. Human)	0.956	— (pending)
p -value	0.0000	—
Paired evaluations (n)	92	92
MCQ Accuracy	34.6%	—
Datasets	K-QA (201) + USMLE (54 MCQ)	same
SLMs evaluated	6 models	6 models
LLM Judge	DeepSeek-v4-flash	same

Note

RAG before/after delta values ($\Delta\rho$) will be populated after full pipeline execution is completed and human evaluations are consolidated.

6 Error Taxonomy — Quality Flags

Quality flags annotate specific failure modes in model responses. They enable structured error analysis and help explain systematic disagreements between judge and human evaluators. Table 4 summarises all six flags.

Table 4: Quality flag taxonomy (inherited from Activity 2).

Flag	Definition and Judge Impact
PROMPT_LEAKAGE	Model echoed or paraphrased system/user prompt verbatim instead of reasoning independently. <i>Impact</i> : inflates perceived quality; judge may award high scores to content-empty answers.
HALLUCINATION	Model fabricated medical facts, drug names, dosages, or clinical guidelines. <i>Impact</i> : critical in medical NLP; may cause judge overrating if the fabrication is linguistically fluent.
OVER_VERBOSE	Response excessively long, padded with filler or redundant caveats. <i>Impact</i> : strongly linked to judge overrating; LLMs may conflate length with quality.
FACTUAL_ERROR	Response contains a verifiably incorrect statement (wrong diagnosis, contraindicated treatment, wrong drug class). <i>Impact</i> : dangerous in clinical settings; occasionally missed by LLM judges.
PARTIAL_ANSWER	Response addresses only part of the question, omitting key clinical components. <i>Impact</i> : common source of underrating; judges may score generously if the answered portion is correct.
OFF_TOPIC	Response does not address the clinical question; pivots to a different condition or gives generic information. <i>Impact</i> : typically produces strong disagreement; human reviewers detect irrelevance more reliably than LLM judges.

7 Judge Reliability — Statistical Interpretation

7.1 Spearman Rank Correlation

Spearman’s ρ measures the degree of rank-order agreement between judge scores and human average scores. Unlike Pearson’s r , it is robust to outliers and does not assume a linear relationship — making it appropriate for ordinal evaluation scales (1–5).

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad \text{where } d_i = \text{rank}(H_i) - \text{rank}(J_i) \quad (3)$$

A value of $\rho = 1.0$ indicates perfect rank agreement; $\rho = 0.0$ means no correlation; $\rho = -1.0$ means a complete inversion of rankings.

7.2 Interpretation Reference

Table 5: Spearman ρ interpretation ranges.

ρ Range	Interpretation	Recommended Action
0.90 – 1.00	Very Strong Agreement	High reliability; ready for deployment.
0.70 – 0.89	Strong Agreement	Minor rank differences; judge reliable.
0.50 – 0.69	Moderate Agreement	Systematic differences; review rubric.
0.30 – 0.49	Weak Agreement	Significant bias; investigate disagreements.
0.00 – 0.29	Very Weak Agreement	Rankings largely uncorrelated.
< 0.00	Disagreement	Judge inverts human rankings; severe issue.

7.3 Bias Metrics

$$\text{Overrating Rate} = \frac{|\{i : J_i > H_i\}|}{N} \quad (\text{concerning if } > 40\%) \quad (4)$$

$$\text{Underrating Rate} = \frac{|\{i : J_i < H_i\}|}{N} \quad (\text{concerning if } > 40\%) \quad (5)$$

$$\bar{\Delta} = \frac{\sum_i |J_i - H_i|}{N} \quad (\text{needs review if } \bar{\Delta} > 1.5) \quad (6)$$

Spearman ρ should be interpreted cautiously when $N < 10$. With fewer than 3 paired evaluations, correlation cannot be computed. Aim for at least 20–30 human-evaluated responses per judge model for robust reliability estimates.

7.4 Python Implementation

```

1 from scipy.stats import spearmanr
2
3 rho_base, p_base = spearmanr(judge_scores_baseline, human_scores)
4 rho_rag, p_rag = spearmanr(judge_scores_rag, human_scores)
5
6 print(f"Baseline rho = {rho_base:.3f} (p = {p_base:.4f})")
7 print(f"With RAG rho = {rho_rag:.3f} (p = {p_rag:.4f})")
8 print(f"Delta rho = {rho_rag - rho_base:+.3f}")

```

Listing 1: Spearman correlation before/after RAG (Python).

8 Database Schema

Activity 3 extends the Activity 2 schema with the following additions.

8.1 New Tables

```

1 CREATE TABLE rag_documents (
2     id          SERIAL PRIMARY KEY,
3     title       TEXT NOT NULL,
4     source_url  TEXT,
5     pub_date    DATE,
6     scope       VARCHAR(20) DEFAULT 'both', -- 'mcq' / 'open' / 'both'
7     created_at  TIMESTAMP DEFAULT NOW()
8 );
9
10 CREATE TABLE rag_chunks (
11     id          SERIAL PRIMARY KEY,
12     document_id INTEGER REFERENCES rag_documents(id),
13     chunk_index INTEGER,
14     content     TEXT NOT NULL,
15     embedding    vector(1536)           -- pgvector
16 );

```

Listing 2: New tables: rag_documents and rag_chunks.

8.2 Schema Extensions

```

1 -- model_responses
2 ALTER TABLE model_responses
3     ADD COLUMN rag_enabled BOOLEAN DEFAULT FALSE,
4     ADD COLUMN rag_context TEXT;
5
6 -- judge_evaluations
7 ALTER TABLE judge_evaluations
8     ADD COLUMN score_with_rag INTEGER,
9     ADD COLUMN rag_impact     VARCHAR(20),
10    -- 'RAG HELPED' / 'RAG ADDED NOISE' / 'RAG NEUTRAL'
11    ADD COLUMN cot_rag        TEXT;

```

Listing 3: Extensions to model_responses and judge_evaluations.

8.3 Key Before/After Comparison Query

```

1 SELECT
2     m.model_name,
3     d.dataset_name,
4     ROUND(AVG(je.score)::numeric, 3)          AS avg_score_baseline,
5     ROUND(AVG(je.score_with_rag)::numeric, 3) AS avg_score_rag,
6     ROUND((AVG(je.score_with_rag)
7         - AVG(je.score))::numeric, 3)          AS delta,
8     COUNT(*) FILTER (WHERE je.rag_impact = 'RAG HELPED')
9         AS rag_helped,
10    COUNT(*) FILTER (WHERE je.rag_impact = 'RAG ADDED NOISE')
11        AS rag_noise,
12    COUNT(*) FILTER (WHERE je.rag_impact = 'RAG NEUTRAL')
13        AS rag_neutral,
14    COUNT(*) AS total
15 FROM judge_evaluations je
16 JOIN model_responses mr ON je.response_id = mr.id
17 JOIN models m          ON mr.model_id    = m.id
18 JOIN questions q       ON mr.question_id = q.id

```

```

19 JOIN datasets d          ON q.dataset_id = d.id
20 WHERE je.score_with_rag IS NOT NULL
21 GROUP BY m.model_name, d.dataset_name
22 ORDER BY m.model_name, d.dataset_name;

```

Listing 4: SQL query for before/after RAG comparison per model and dataset.

9 Repository Structure

```

MedEval-RAG-as-a-Judge-Framework-Atividade-3/
|-- README.md
|-- Atividade_3/
|   |-- documents/                # Source PDFs/TXTs for RAG
|   |-- scripts/
|       |-- 01_ingest_documents.py # Chunking + pgvector indexing
|       |-- 02_rag_inference.py    # RAG-augmented model inference
|       |-- 03_judge_rag.py        # LLM Judge with Chain-of-Thought
|       |-- 04_update_db.py        # PostgreSQL schema update + insert
|       |-- 05_statistics.py       # Spearman rho before/after
|   analysis
|       |-- prompts/
|           |-- rag_system_prompt.txt # Augmented inference prompt
|   template
|       |-- judge_rag_prompt.txt     # Judge prompt with RAG impact
|   field
|       |-- notebooks/
|           |-- rag_pipeline_demo.ipynb # End-to-end pipeline walkthrough
|           |-- statistics_analysis.ipynb
|       |-- backup/
|           |-- activity3_before_after.sql # PostgreSQL dump (pre + post
|   RAG)
|       |-- exports/
|           |-- spearman_results.csv    # rho before/after per model/
|   dataset
|       |-- rag_noise_cases.csv        # Cases where RAG degraded
|   performance
|-- artifacts/                        # (inherited from Activity 2)
|   |-- api-server/
|   |-- llm-judge/
|-- lib/
|   |-- db/src/schema/
|       |-- rag.ts                    # rag_documents + rag_chunks tables

```

Listing 5: Repository directory layout.

10 Technology Stack

Table 6: Technology stack.

Layer	Technology
Frontend	React 19, Vite, TypeScript, Tailwind CSS, Shadcn UI
Backend	Express.js, Node.js, TypeScript
Database	PostgreSQL + Drizzle ORM
Vector Search	pgvector (1536-dim embeddings)
Embeddings	OpenAI <code>text-embedding-3-small</code>
LLM Providers	OpenAI, Google Gemini, Anthropic Claude, DeepSeek
Auth	Firebase Admin + Clerk
Monorepo	pnpm workspaces

11 Setup and Deployment

11.1 Prerequisites

- Node.js 20+
- pnpm 9+
- PostgreSQL 14+ with pgvector extension
- OpenAI API key (embeddings + judge)

11.2 Installation

```
# Install dependencies
pnpm install

# Enable pgvector
psql -c "CREATE EXTENSION IF NOT EXISTS vector;"

# Run API server
pnpm --filter @workspace/api-server run dev

# Run frontend
pnpm --filter @workspace/llm-judge run dev
```

Listing 6: Installation and development server startup.

11.3 Database Restore

```
# 1. Create database
createdb -U postgres medeval_db

# 2. Restore Activity 2 baseline
psql -U postgres -d medeval_db \
  -f Atividade_2/backup/medeval-backup-2026-05-14.sql

# 3. Apply Activity 3 schema extensions
```

```
psql -U postgres -d medeval_db \
-f Atividade_3/backup/activity3_before_after.sql

# 4. Verify
SELECT COUNT(*) FROM rag_documents;
SELECT COUNT(*) FROM rag_chunks;
SELECT COUNT(*) FROM judge_evaluations WHERE score_with_rag IS NOT NULL;
```

Listing 7: Restoring the full Activity 2 + Activity 3 database.

12 Security Notes

- API keys are stored privately per user and deleted on logout — never included in backups.
- All evaluations are authenticated via JWT.
- Backup files contain evaluation data only — they must be stored securely.

13 Terminology Reference

Table 7: Key terminology used throughout the report.

Term	Definition
Gold Answer	Authoritative reference answer from the original dataset. True ground truth.
LLM / Synthetic Ref.	Answer generated by a large Judge LLM to guide scoring. NOT a ground truth.
SLM (Candidate)	Small Language Model being evaluated.
Judge Model	LLM used to score SLM responses.
LLM-as-a-Judge	Evaluation paradigm where an LLM judges another model’s outputs.
Human Baseline	Human-assigned scores used as calibration reference for judge reliability.
Spearman ρ	Rank-order correlation between judge and human scores. Range $[-1, +1]$.
Disagreement Δ	$ \overline{H} - J $. A delta ≥ 2 triggers disagreement analysis.
OVERRATING	Judge assigns higher score than humans. Associated with leniency bias.
UNDERRATING	Judge assigns lower score than humans. Associated with strictness/paraphrase failure.
Quality Flag	Human-assigned label identifying a specific failure mode in a response.

References

- [1] Zheng, L., et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. arXiv:2306.05685, 2023.

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Federal University of Sergipe

Department of Computing

Special Topics in Software Engineering and Information Systems I

MedEval

*LLM-as-a-Judge Evaluation Platform
with Human Alignment*

Activity 2 — Group 5

Member	Main Contribution
Moaath Alshaikh	Full-stack development, database architecture, Spearman analysis
Tasneem Alshaher	SLM models, dataset management, prompt engineering
Marcelo West	Database architecture
Clélio Xavier	Evaluation pipeline (judge)
Sérgio Santos	Human evaluation, results analysis
Hernandison Bispo	Research and insight generation

Supervisor: Prof. Glauco Carneiro

Date: May 2026

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1. Overview

The **MedEval** platform is a research system that implements the *LLM-as-a-Judge* paradigm to evaluate Small Language Models (*Small Language Models* — SLMs) in medical question-answering tasks. The platform stores the entire evaluation lifecycle in a PostgreSQL relational database, ensuring complete traceability and reproducibility.

“It is not enough to evaluate — it is necessary to know who evaluated, when, and under which explanatory rationale.”

— Prof. Glauco Carneiro

The platform was designed and developed by **Moaath Almohammad Alshaikh**, who built the full-stack system — React frontend, Node.js backend, and PostgreSQL integration — based on the initial database schema established by **Marcelo West**.

1.1. Project Repository

The complete source code, evaluation scripts, data exports, and documentation are available in the public GitHub repository:

<https://github.com/moaathalshaikh/MedEval-LLM-as-a-Judge-Framework-Atividade-2>

The repository includes a complete PostgreSQL database backup, datasets exported in JSONL and CSV formats, Python scripts for Spearman correlation analysis, the LLM judge pipeline, and the prompts used during evaluations.

2. Main Results

Table 1 summarizes the key metrics obtained during the complete execution of the evaluation pipeline.

Tabela 1: Main experiment metrics

Metric	Value
Spearman Correlation ρ (Judge vs. Human)	0,956
p -value	0,0000
Paired evaluations (n)	92
MCQ Accuracy	34,6%
Datasets used	K-QA (201 open-ended) + USMLE (54 MCQ)
Evaluated SLMs	6 models
LLM Judge	DeepSeek-v4-flash

3. Evaluation Pipeline

The complete evaluation workflow consists of five sequential stages, as illustrated below:

1. **Dataset** (K-QA / USMLE) — open-ended and multiple-choice questions in the medical domain.

2. **SLM Response Generation** — 6 candidate models (originating from Activity 1).
3. **Reference Answer Generation** — DeepSeek-v4-flash generates a synthetic reference answer for open-ended questions.
4. **LLM Judge Evaluation** — score from 1–5 with structured reasoning for each response.
5. **Human Review** — independent scores assigned by the project team members.
6. **Spearman Correlation ρ** — measures the agreement level between the LLM Judge and human evaluators.
7. **Research Analysis** — bias detection, error taxonomy, and disagreement analysis.

4. Spearman Correlation

We used Spearman rank correlation to measure the degree of agreement between the LLM Judge and human evaluators:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (1)$$

where d_i is the difference between the judge score and the human score for each evaluated response.

The code snippet below demonstrates the calculation performed by the `spearman_analysis.py` script:

```

1 from scipy.stats import spearmanr
2
3 correlation, p_value = spearmanr(human_scores, judge_scores)
4 print(f"Spearman rho = {correlation:.3f}, p = {p_value:.4f}")
5 # Output: Spearman rho = 0.956, p = 0.0000

```

Listing 1: Spearman correlation calculation

Table 2 presents the interpretation of ρ intervals:

Tabela 2: Interpretation of Spearman correlation

Range of ρ	Interpretation
$0,7 \leq \rho \leq 1,0$	Strong alignment ✓
$0,3 \leq \rho \leq 0,6$	Moderate — review rubric
$\rho < 0,3$	Weak judge

The obtained result ($\rho = 0,956$) falls within the **strong alignment** range, indicating that the LLM Judge is highly consistent with human judgment in this domain.

5. Scoring Rubric

5.1. Open-Ended Questions

Table 3 describes the rubric used to evaluate responses to open-ended questions.

Tabela 3: Scoring rubric for open-ended questions

Score	Label	Description
5	Excellent	Equal to or better than the LLM reference answer
4	Good	Clinically correct, with minor omissions
3	Partial	Acceptable, but lacks sufficient precision
2	Weak	Serious clinical omission
1	Critical	Hallucination or clinically dangerous error

5.2. Multiple-Choice Questions (MCQ)

MCQ questions are evaluated deterministically: the letter selected by the model is compared against the gold answer, without requiring an API call to the Judge.

6. Error Flag Taxonomy

During qualitative analysis, problematic responses were tagged with *flags* according to Table 4.

Tabela 4: Quality flag taxonomy

Flag	Meaning
PROMPT_LEAKAGE	Model repeated prompt instructions
HALLUCINATION	Fabricated clinical information
OVER_VERBOSE	Excessively long answer that influences the judge
FACTUAL_ERROR	Medically incorrect statement
PARTIAL_ANSWER	Incomplete response
OFF_TOPIC	Irrelevant content

7. Repository Structure

```

1 Atividade_2/
2 |-- README.md
3 |-- backup/
4 |   |-- medeval-backup-2026-05-14.sql
5 |   +-- medeval-backup-2026-05-14.sql.sha256
6 |-- exports/
7 |   |-- medeval-research-2026-05-14.jsonl
8 |   +-- medeval-research-2026-05-14.csv
9 |-- scripts/
10 |   |-- spearman_analysis.py
11 |   |-- import_responses.py

```

```
12 |   +-- run_judge.py
13 | -- prompts/
14 |   |-- judge_medical_open.txt
15 |   |-- judge_medical_mcq.txt
16 |   +-- reference_answer.txt
17 | -- sql/
18 |   +-- schema.sql
19 | -- tutorial/
20 |   +-- restore_tutorial.pdf
21 +-- presentation/
22   +-- activity2_presentation.pdf
```

Listing 2: Repository directory structure

8. Database Restoration

8.1. Requirements

- PostgreSQL 14 or higher
- psql CLI

8.2. Steps

```
1 # 1. Create the database
2 createdb -U postgres medeval_db
3
4 # 2. (Optional) Verify backup integrity
5 sha256sum -c medeval-backup-2026-05-14.sql.sha256
6
7 # 3. Restore
8 psql -U postgres -d medeval_db \
9     -f backup/medeval-backup-2026-05-14.sql
10
11 # 4. Verify record counts
12 psql -U postgres -d medeval_db -c \
13     "SELECT COUNT(*) FROM questions_w_answers;"
14 # Should return: 255
15
16 psql -U postgres -d medeval_db -c \
17     "SELECT COUNT(*) FROM judge_evaluations;"
18 # Should return: 92+
```

Listing 3: PostgreSQL database restoration

Detailed instructions with screenshots are available in `tutorial/restore_tutorial.pdf`.

9. Research Data Export

Two export formats are available in the `exports/` directory:

- **JSONL** (`medeval-research-2026-05-14.jsonl`) — for ML pipelines and Python scripts.
- **CSV** (`medeval-research-2026-05-14.csv`) — for Excel, SPSS, and statistical tools.

The exports contain only research data: questions, SLM responses, judge scores, human scores, and flags. Operational data (users, logs, API keys) are excluded.

10. Security

- API keys are stored privately per user and are **deleted upon logout** — never included in backups.
- All evaluations are authenticated using **JWT**.
- Backup files contain only evaluation data — store them securely.

11. References

- [1] Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. arXiv:2306.05685, 2023.
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